A balance-consistent restart for the moving-cavity mesh change

Deriving the conditions newly-opened and newly-iced cells must satisfy for settle-free continuation (rev. 2, post-review)

AWI-ESM3 ↔ PISM moving cavity — design note

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Abstract

On every moving-cavity leg the FESOM restart is remapped onto a sub-mesh whose ice front has moved. Empirically the leg then blows up within days: *fast* as a vertical-CFL runaway at $\Delta t = 1200\,\mathrm{s}$, or *slow* as a barotropic $\eta_n \to$NaN at $\Delta t = 60\,\mathrm{s}$. We have falsified every *value-only* restart fix (removing static density inversions; nearest-neighbour velocity; running without a settle). This note shows the two blow-ups are two faces of a single **geostrophic adjustment**, that the balanced state a stable restart must be in is determined by the pressure field (density + ssh + FESOM's cavity pressure boundary condition) together with the surrounding spun-up water masses, and that the required velocity can therefore be **computed** per level from $\nabla_h p$ — one formula, no reference ambiguity. The plan removes the *leading-order* adjustment by construction; the residual discrete imbalance is *measured* before the run by explicit balance/stability metrics (§??, step V).

1 Status: what is falsified

The instability is a coastal, mid-column ($\sim$95 m) mode that grows over days from the changed margin. Ruled out, each verified at the data level and each leaving the blow-up essentially unchanged:

- **Static density inversions** — a coherent-column T/S refill removed all severe inversions; the leg still blew (`mstep`≈164, CFLz≈8.5). *Metric (for reproducibility):* linearised EOS $\rho = 1027\,(1-1.7\times10^{-4}T+7.6\times10^{-4}(S-35))$, adjacent-level $\Delta\rho$ over *wet levels only* (`ulevels`..`nlevels`-1), minimum per column, over the 1595 retreated cells of the movcav8 chunk-3 sub-mesh: worst $-8.8 \to -0.5\,\mathrm{kg\,m^{-3}}$, cells with $\Delta\rho < -2$: $11 \to 0$. (Including dry sub-bottom padding or cavity levels above `ulevels` yields spurious ~ -27 values from S=0/melt outliers — do not compare against unmasked recomputes.)
- **Nearest-neighbour velocity** (Finn Heukamp / SO-ASE handling) — filling changed elements with the nearest old element's current (no cell left at zero) lowered peak CFLz $8.5 \to 5.7$ but *still* blew at `mstep`≈164, now with a barotropic $\eta_n \to$NaN as well: the copied current is not balanced against the local density *and* its depth-integral is not transport-consistent with η (§??).
- **No settle / timestep** — $\Delta t = 60\,\mathrm{s}$ removed the fast CFLz mode entirely (0 warnings, $\sim$10 days) but a slow barotropic η_n mode grew and NaN'd.

A value-only restart cannot be made *balanced*: the imbalance is lateral (fronts) + missing momentum, not a vertical inversion.

2 FESOM's cavity pressure convention (pinned from source)

Before writing balance conditions, the ssh/ice-load convention must be fixed; from `oce_ale_pressure_bv.F90` (`pressure_force_4_linfs_cavity` path, active for `use_cavity`):

- The ice-shelf load is **not** $\rho_i g h_i$. FESOM fills the cavity-occupied levels 1..`ulevels`−1 with a **virtual water column carrying the ice-base** T, S (density re-evaluated at each level's pressure) and integrates it hydrostatically to get the pressure BC at the ice base. For a homogeneous ocean this construction is exactly pressure-gradient-free across the front (the code comment states this design goal).
- η is **0 under the cavity** (verified in the evolved mother restart: all 1591 sub-ice nodes carry $\eta = 0.000$ exactly); the open-ocean η is prognostic.

Hence the correct hydrostatic pressure at level z, everywhere, is

$$p(z) \;=\; \rho_0\, g\, \eta + g\int_z^0 \tilde{\rho}\, dz', \qquad \tilde{\rho} = \begin{cases} \rho(T,S,z') & \text{in the wet column,} \\ \rho(T_b,S_b,z') & \text{in cavity-occupied levels } (T_b, S_b: \text{ ice-base } T, S), \end{cases} \tag{1}$$

with η=0 under ice. This replaces the $\rho_i g h_i$ form of rev. 1 (which was inconsistent with the barotropic term there) and automatically handles the newly-iced margin: the ice load is carried by the virtual column, so no separate ice-slope term is needed anywhere.

3 The balance a stable restart must satisfy

"No adjustment shock" is equivalent to the restart satisfying, at every wet point:

$$\text{(1) static stability:}\quad N^2 = -\frac{g}{\rho_0}\,\partial_z\rho \;\geq\; 0, \tag{2}$$

$$\text{(2) hydrostatic pressure:}\quad p(z) \text{ per Eq. (??)}, \tag{3}$$

$$\text{(3) geostrophy, per level:}\quad \mathbf{u}_g(z) = \frac{1}{f\rho_0}\,\hat{\mathbf{z}}\times\nabla_h p(z), \tag{4}$$

$$\text{(4) transport consistency:}\quad \partial_t\eta = -\nabla_h\cdot\!\int \mathbf{u}\, dz \approx 0, \tag{5}$$

$$\text{(5) ALE consistency:}\quad \textstyle\sum_k \texttt{hnode}_k = D + \eta \quad (\text{no spurious } \partial_t h \to w). \tag{6}$$

Eq. (3) is **one formula per level** — thermal wind and the barotropic reference are both contained in $\nabla_h p$, so there is no separate "integration constant" to choose and no neighbour-velocity reference to add (rev. 1's step-3/step-4 split double-counted the barotropic part; this supersedes it). (1) keeps convection off; (4) is what gates the *slow* barotropic η_n blow-up and must be checked on the *depth-integrated* field, not just pointwise; (5) is a necessary condition that the *current remap violates* (`remap_hnode` assigns nominal z-level thicknesses ignoring η) and therefore gets its own implementation step.

Continuous vs discrete: geostrophic flow is non-divergent in the continuum; the entire residual risk is the *discrete* divergence at the changed front. The claim is therefore "removes the leading-order adjustment", with the residual measured (step V), not "balanced by construction".

4 Quantitative diagnosis: the missing jet

At the Weddell seed the fill leaves a front of $\Delta S \approx 3\,\text{psu}$ over $\Delta x \approx 25\,\text{km}$: $\Delta\rho \approx 2.4\,\text{kg}\,\text{m}^{-3}$, $\nabla_h\rho \approx 1.0\times10^{-4}\,\text{kg}\,\text{m}^{-4}$, and with $f(-66°) \approx -1.33\times10^{-4}\,\text{s}^{-1}$, $\partial_z u = \frac{g}{f\rho_0}\nabla_h\rho \approx 7\times10^{-3}\,\text{s}^{-1} \Rightarrow \Delta u \approx 0.7\,\text{m}\,\text{s}^{-1}$ over 100 m. **The balanced front carries a $\sim$0.7 m s^{-1} jet**; we initialised it at 0 (v4) or $\sim$0.06 m s^{-1} (v5) — an $O(0.7)$ momentum deficit, which is the observed violent adjustment. NN cannot supply it: the required current is specific to *this* density gradient.

5 Conditions per case

Added points (ice retreated / newly wet). At each newly-wet level: (i) T, S laterally *anchored to the neighbouring water mass at that level* and vertically statically stable — the blend must be **density-monotonic** (enforce $N^2 \geq 0$ jointly on T, S, e.g. by convective adjustment), *not* independent T, S interpolation, since mixing fresh ice-base water below denser open-ocean water manufactures inversions; (ii) $\mathbf{u}$ from Eq. (3); (iii) η from the open-ocean neighbours (0 if still sub-ice); (iv) `hnode` per (5).

Points next to removed points (ice advanced / newly iced). With the pinned convention (§**??**) the new ice load is carried by the virtual-column pressure BC; the adjacent open cells need no special term — their Eq. (3) velocity, computed from Eq. (**??**), already contains the geostrophic response to the new load.

6 Corollary: do not over-sharpen the front

A $0.7\,\mathrm{m\,s^{-1}}$ jet is high for an Antarctic cavity front, i.e. the current fill over-sharpens it. *Scope (corrected):* fully surface-opened columns already use a coherent open-ocean donor; the "hold shelf-base water upward" path fires only for **still-sub-ice thinned columns** — which is exactly the observed seed (ulev $14 \to 2$, $S \approx 31.0$ through the column). The fix and its target are therefore: fill each newly-wet level of those columns from the **nearest old node wet at that level** (lateral anchoring — by construction the front gradient matches the neighbouring unchanged cells, removing the free "softening knob"), then enforce $N^2 \geq 0$ column-wise (density-monotonic adjustment). Gentler front $\Rightarrow$ smaller required jet $\Rightarrow$ smaller residual after step (3).

7 Transport consistency (the slow mode)

$\partial_t \eta = -\nabla_h \cdot \int \mathbf{u}\, dz$: the slow η_n blow-up is fed by any mismatch between the depth-integrated velocity and η. Computing $\mathbf{u}$ per level from $\nabla_h p$ makes $\int \mathbf{u}\, dz$ automatically the geostrophic transport of the *same* p — i.e. transport consistency holds to discretisation error. The verification therefore checks $|\nabla_h \cdot (H\bar{\mathbf{u}})|$ explicitly (step V), which is the quantity that gates the slow mode; pointwise $|\nabla_h \cdot \mathbf{u}|$ alone is not sufficient.

8 Implementation plan (`mo_remap_fields.F90` unless noted)

I. **`hnode`/η ALE consistency (new; cheapest, do first).** In `remap_hnode`, scale the nominal thicknesses of changed columns so $\sum_k h_k = D + \eta$ (zstar distribution), with η the staged (donor-filled, or 0 sub-ice) value. Removes a standing $\partial_t h / \partial_t \eta$ source at t=0; a leading suspect for the slow mode.

II. **Laterally-anchored, density-monotonic T/S for still-sub-ice thinned columns.** Replace "hold shelf-base upward" by the per-level nearest-wet-old-node donor (memoised cache already exists), then a joint T, S convective-adjustment pass over every changed column (also the donor-filled fully-opened ones) to guarantee $N^2 \geq 0$.

III. **Geostrophic velocity on changed elements.** Post-pass over the written new-mesh restart: build $p(z)$ per node via Eq. (**??**) (linearised EOS; virtual column with ice-base T, S under cavities; $+\rho_0 g \eta$, η=0 sub-ice); per-level P_1 gradient on each changed triangle; $\mathbf{u}_g = \frac{1}{f\rho_0} \hat{\mathbf{z}} \times \nabla_h p$. Overwrite `u,v` there. No NN reference anywhere (supersedes v5; the NN fill remains only as a fallback where $|f|$ is degenerate, which does not occur on the Antarctic margin).

IV. Zero the Adams–Bashforth history on changed elements (`urhs_AB,vrhs_AB`): a balanced $\mathbf{u}$ with a foreign AB tendency reintroduces a 1–2-step transient; zeroing gives a clean AB restart (FESOM handles first-step AB init).

V. Verify *before* running (offline, on the staged restart, changed cells): (a) $\min N^2$; (b) $\max|\nabla_h\cdot(H\bar{\mathbf{u}})|$ *and* pointwise $|\nabla_h\cdot\mathbf{u}|$; (c) thermal-wind residual $|\partial_z\mathbf{u}-\frac{1}{f\rho_0}\hat{\mathbf{z}}\times\nabla_h\partial_z p|$; (d) **stability of the constructed field**: advective CFL $|\mathbf{u}|\Delta t/\Delta x$, vertical shear Richardson number $Ri=N^2/|\partial_z\mathbf{u}|^2$ (flag $Ri<0.25$), and $\max|\mathbf{u}|$ sanity. The field must be both *balanced* and *integrable*.

VI. Re-test the movcav8 chunk-3 leg at 1200 s, no settle.

9 Caveats / open points

- **Residual discrete divergence** at the changed front is the remaining risk; it is measured (step V-b) rather than assumed away.
- **Ageostrophic background** (Ekman/tides) is not reconstructed; for a spun-up interior geostrophy dominates at t=0, and unchanged cells keep their full evolved velocity.
- **Forcing transient:** a balanced restart is steady only under steady forcing; the atmosphere will immediately flux the newly-opened cells. Settle-free means "no adjustment shock", not "forcing-free".
- **Still-sub-ice retreat cells** (the actual seed) get their T/S from cavity-side per-level donors and $\eta=0$; their velocity still comes from Eq. (3) — the virtual column makes the pressure well-defined there.
- **Source term:** if the per-leg front motion is itself implausibly large (deep-draft transitions in one 10-yr PISM step), gentling the `cavity_nlvls` regeneration reduces $\nabla_h\rho$ at the root and makes every step above easier.